

OLIVIA WANG

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EDUCATION

California Institute of Technology Pasadena, CA | GPA: 4.0/4.0

Jun 2027

B.S. in Computer Science

- Courses: Machine Learning, Deep Learning, Software Engineering, Operating Systems, Data Structures, Algorithms, Probability, Statistics, Linear Algebra, Discrete Math, Economics, Law & Finance for Startups

EXPERIENCE

Software Development Intern | Amazon Music Culver City, CA

Incoming Jun 2026

Machine Learning Researcher | Perona Vision Lab, Caltech

Pasadena, CA

Jan 2025 - Present

- Architected PyTorch-based infrastructure to benchmark vision-language diffusion, autoregressive models against human reasoning datasets across 10+ visual tasks.
- Developed a high-precision human-study application for 100 users using Python & Javascript for web-based data collection with precise stimuli timing.
- Engineered Python backends to optimized GPU system performance for batch processing, distributed data parallelization, and concurrent execution on HPC.

Software Engineering Intern | Eli Lilly's Protomer Technologies Pasadena, CA

Jan 2025 - Present

- Architected and deployed a full-stack web application using Python (Dash) to visualize high-frequency glucose sensor data, enabling hardware-in-the-loop experiments for critical medical R&D.
- Built a multi-threaded backend in Python to ingest BLE sensor data and deliver real-time ML model predictions for automated insulin regulation logic.

Computational Simulation Researcher | Murray Lab, Caltech

Pasadena, CA

May 2022 - Jun 2025

- Engineered Python pipeline for parameter sweeps, visualization, and data export to improve protein expression efficiency by 2x in modular biochemical system models with BioCRNpyler.
- Awarded Doris S. Perpall Speaking Competition Semifinalist 2024.

PUBLICATIONS

Wang, O., et al. "[Vision Language Model Comparison with Humans]." Submitted to ECCV 2026.

Wang, O., Larsson, E., Murray, R. "A DNA Part Library for *X. griffinae*." *ACS Synthetic Biology*. (2025).

PROJECTS

Sheet Engine with Automated Logic Validation | Python, Pytest, Unittest, API

May 2025 - Jun 2025

- Led a 3-person team to engineer a spreadsheet calculation engine with complex formula parsing and dependency tracking, optimizing for speed and scale.
- Developed automated unit & integrated testing suite, ensuring 100% calculation accuracy and system reliability across edge cases.

Scalable Deep Learning for Radiograph Diagnostics | PyTorch, SLURM, CUDA

Apr 2025 - Jun 2025

- Optimized PyTorch model training (DenseNet) of 60,000+ patients' data on HPC clusters (SLURM) using distributed data parallelization to predict 9 pathologies with an average 0.78 MSE.

SKILLS

- **Software**: Python, Java, C, Javascript, PostgreSQL, Git, Linux, Containerization, React, Automated Tesing
- **Machine Learning**: PyTorch, Tensorflow, scikit-learn, Deep Learning, Computer Vision, LLM, NLP, AWS

LEADERSHIP

- Machine Learning Teaching Assistant for 100+ students, Caltech Jan 2026 - Present
- Health Advocate, Social Media & Admissions Ambassador, Caltech May 2024 - Present
- Co-founder, President, Harvard-Westlake Science Olympiad Aug 2020 - Jun 2023
- Team USA and National Figure Skater, Los Angeles Ice Theater Sept 2018 - Jun 2024